

# REX REMIGIUS STEPHEN JOTHI

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## EDUCATION

**Northeastern University, Boston, MA**

2025 - Expected 2027

Master of Science in Computer Science

**Relevant Course Work :** Programming Design Paradigm, Computer Systems

**Sri Krishna College of Engineering and Technology, Coimbatore, India**

2019 - 2023

Bachelor of Technology in Information Technology

(GPA: 8.68/10.00)

**Relevant Course Work:** Object Oriented Programming, Computer Architecture, Data Structures and Algorithms, Machine Learning Techniques and Big Data Analytics

## SKILLS

**Programming Languages:**

C, C++, Python, Java

**Compiler & Systems:**

LLVM, GCC, Intrinsic Programming, SIMD Optimization

**Machine Learning & AI Frameworks:**

TensorFlow, PyTorch, ONNX, TFLite, NumPy

**Development & Testing Tools:**

Git/GitHub, PyTest, LIT, Docker, VSCode, Jupyter

## WORK EXPERIENCE

**Software Engineer, MulticoreWare Inc**

Jan 2023 - Jul 2025

- Designed and implemented custom compiler intrinsics for LLVM and GCC, enabling code reuse across projects and improving execution efficiency by 15% through advanced optimization techniques and memory management strategies.
- Developed and integrated a novel optimization algorithm into LLVM's code generation pipeline, achieving a 5% performance improvement in generated binaries through enhanced instruction scheduling and register allocation.
- Architected and optimized machine learning kernels (convolution, maxpool, normalization) for a custom AI accelerator, reducing inference latency by 63% for single-stream workloads and 67% for multi-stream processing through efficient memory access patterns and SIMD optimization.
- Benchmarked and profiled multiple ML models (CNNs, transformers) on the AI accelerator architecture, identifying performance bottlenecks and implementing targeted optimizations that improved throughput by 40%.
- Awarded Employee of the Month twice for designing high-performance ML kernels for the AI accelerator and successfully migrating an audio processing model from PyTorch to TensorFlow and TFLite with zero accuracy loss.

**Summer Intern, Zoho Corporation**

Apr 2022 - May 2022

- Developed a Java command-line application with optimized data structures and algorithms, achieving 25% runtime performance improvement through efficient memory management and algorithmic optimization.
- Solved 100+ algorithmic problems across data structures, OOP design patterns, and the Java Collections Framework, strengthening debugging skills and understanding of computational complexity.

## PROJECTS

**Enhanced Address Sanitizer pass in LLVM for fault tolerance system**

C, LLVM, C++, Shell, Make

- Enhanced LLVM's Address Sanitizer with fault-tolerant execution by implementing **CIMA (Countering Illegal Memory Access)** algorithm and thereby, achieving **100%** runtime availability, 0% crash rate, and **30%** lower reallocation overhead through custom CFG-based instrumentation and adaptive memory resizing.
- Integrated 3 new LLVM passes and plugin layers, improving error reporting and achieving **100%** detection accuracy across 20+ verified test cases covering heap, stack, and global memory errors.

**Transparent Checkpoint and Restart System**

C, POSIX signals, Linux syscalls, GDB

- Engineered transparent checkpoint-restart system—core technology in Docker, Kubernetes, and CRIU—that captured **full process state (memory + CPU registers)** and restored execution to identical state in new address space without modifying target code, enabling fault tolerance and seamless process migration.
- Solved critical systems challenges - address space conflict resolution through static compilation, call frame consistency management, and register restoration via assembly-level debugging.

**Heart Disease Risk Prediction Model**

Python, Pandas, NumPy, Matplotlib, Scikit-learn

- Achieved **90%+** prediction accuracy (**94% ROC-AUC**) on UCI Heart Disease dataset by implementing ensemble learning methods that outperformed baseline logistic regression by 6%. Optimized model performance through **960+ hyperparameter combinations** via Grid Search with 10-fold cross-validation, achieving balanced precision (89%) and recall (91%).